99.500 Applied Mathematics for Engineering

Homework 1 Solution

February 4, 2026

1. (a) By using the shell balance in an elementary slab from $x = x$ to $x = x + \Delta x$, the energy balance can be expressed as

$$\rho c A \Delta x \frac{\partial T}{\partial t} = k \left(\left. \frac{\partial T}{\partial x} \right|_{x+\Delta x} - \left. \frac{\partial T}{\partial x} \right|_{x} \right) A \tag{1}$$

where ρ is slab density, c is the specific heat capacity, A is the cross-sectional area and k is the material conductivity. Dividing both sides by $A\Delta x$ and applying the definition of derivative in the limit $\Delta x \to 0$, the non-steady-state temperature distribution equation for $T(x,t)$ is obtained

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2} \quad \left(\kappa = \frac{k}{\rho c} \right) \tag{2}$$

(b) The initial condition for the whole semi-infinite slab at $t = 0$ is

$$T(x, t = 0) = T_i \tag{3}$$

The boundary condition at surface $x = 0$ is

$$T(x = 0, t) = T_i + At \tag{4}$$

The boundary condition at a position far from the surface is given by

$$T(x \to \infty, t) = T_i \text{ or } \left. \frac{\partial T}{\partial x} \right|_{x \to \infty} = 0 \tag{5}$$

2. (a) By using the steady-state shell balance in the porous bed in the radial direction from $r = r$ to $r = r + \Delta r$ for $r > R$, the mole balance can be expressed as

$$2\pi r L \left. N \right|_{r} = 2\pi r L \left. N \right|_{r+\Delta r} + 2\pi r \Delta r L a r_s \tag{6}$$

where a is the specific interfacial area of the porous medium and r_s is the reaction rate on the surface of the solid particles within the porous medium. Dividing both sides by $2\pi L \Delta r$ and applying the definition of derivative in the limit $\Delta r \to 0$ to obtain

$$-\frac{d}{dr}(rN) = rakC \tag{7}$$

where $r_s = kC$. Substituting the given $N = Cv - D\frac{dC}{dr}$ into (7) to have

$$-\frac{d}{dr}\left[r \left(Cv - D\frac{dC}{dr} \right) \right] = rakC \tag{8}$$

$$Dr\frac{d^2C}{dr^2} + (D - rv)\frac{dC}{dr} - C\frac{d(rv)}{dr} = rakC \tag{9}$$

Assume dilute solution with mass continuity condition $\frac{d(rv)}{dr} = 0$, the equation (9) can be re-arranged into

$$r^2\frac{d^2C}{dr^2} + r\left(1 - \frac{rv}{D}\right)\frac{dC}{dr} - \frac{ak}{D}r^2 C = 0 \tag{10}$$

(b) With no accumulation in the pipe, all the reagent entering through the open end must flow into the porous bed,

$$QC_F = 2\pi RL\ N|_{r=R} = 2\pi RL \left[C(R)v(R) - D\frac{dC}{dr}\bigg|_{r=R} \right] \tag{11}$$

Similarly, the volumetric flow rate can be expressed as

$$Q = 2\pi RLv(R) \tag{12}$$

and substitue equation (12) into (11) to obtain the boundary condition at $r = R$,

$$C(R) - \frac{2\pi RLD}{Q}\frac{dC}{dr}\bigg|_{r=R} = C_F \tag{13}$$

Since the reactant is being consumed, its concentration is expected to decay to zero as r increases. This provides the second boundary condition,

$$C(r \to \infty) = 0 \text{ or } \frac{dC}{dr}\bigg|_{r\to\infty} = 0 \tag{14}$$

(c) From mass continuity condition $\frac{d(rv)}{dr} = 0$, rv is constant and hence $rv = Rv(R)$. With (12) to obtain $rv = \frac{Q}{2\pi L}$ and then substitute this into equation (10),

$$r^2\frac{d^2C}{dr^2} + r\left(1 - \frac{Q}{2\pi LD}\right)\frac{dC}{dr} - \frac{ak}{D}r^2C = 0 \tag{15}$$

Let $\widetilde{C} = \frac{C}{C_F}$ and $\widetilde{r} = \frac{r}{R}$, the equation (15) can be transformed into its dimensionless form as

$$\widetilde{r}^2\frac{d^2\widetilde{C}}{d\widetilde{r}^2} + \widetilde{r}\,(1-\theta)\,\frac{d\widetilde{C}}{d\widetilde{r}} - \phi\widetilde{r}^2\widetilde{C} = 0 \tag{16}$$

where $\theta = \frac{Q}{2\pi LD}$ and $\phi = \frac{akR^2}{D}$. The dimensionless boundary conditons are

$$\widetilde{C}(1) - \frac{2\pi LD}{Q}\frac{d\widetilde{C}}{d\widetilde{r}}\bigg|_{\widetilde{r}=1} = 1 \tag{17}$$

$$\widetilde{C}(\widetilde{r} \to \infty) = 0 \text{ or } \frac{d\widetilde{C}}{d\widetilde{r}}\bigg|_{\widetilde{r}\to\infty} = 0 \tag{18}$$